Department of Computer Science and Engineering

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2023-2027

Sixth Semester Progress Report on

"Decentralized P2P"

Submitted for the partial fulfilment of Project Centric Learning activity of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

Submitted by

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CERTIFICATE

This is to certify that the project centric learning work titled **"Decentralized P2P"** is carried out by **Deepak Kumar Jha (23BTRCN001), Ayadee Aphiwatamorn (23BTRCN009), Abdul Hameed Siddiqui (23BTRCN020), Shaili Srivastava (23BTRCN026), Hanan Zahoor (23BTRCN049),** bonafide students of Bachelor of Technology at the Faculty of Engineering & Technology, Jain Deemed-to-be University, Bangalore in partial fulfillment for the project centric learning activity of degree in Bachelor of Technology in Computer Science & Engineering, during the year **2023-2027**.

Dr D Ramya Dorai	***Dr Mahesh Tr***	***Dr. Geeta***
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ACKNOWLEDGEMENT

It is a great pleasure for us to acknowledge the assistance and support of School of Computer Science and Engineering, Jain (Deemed-to-be University) for the progress of this project centric learning work.

In particular we would like to thank ***Dr Geeta****,* ***Director,*** *School of Computer Science Engineering and* ***Program Dr Mahesh TR, Program Head,*** *Department of Computer Science and Engineering, JAIN (Deemed-to-be University) for their constant encouragement and expert advice.*

We would like to thank our guide ***Dr D ramya Dorai, Department of Computer Science & Engineering****, JAIN (Deemed-to-be University), for sparing his valuable time to extend help in every step of our project work, which paved the way for smooth progress of the project.*

We would like to thank our Project Coordinators ***Prof Mohin Hassan*** *and all the staff members of Computer Science & Engineering for their support.*

We would like to thank one and all who directly or indirectly helped us in the progress of our Project.

Signature of Students

DECLARATION

We, Deepak Kumar Jha (23BTRCN001), Ayadee Aphiwatamorn (23BTRCN009), Abdul Hameed Siddiqui (23BTRCN020), Shaili Srivastava (23BTRCN026), Hanan Zahoor (23BTRCN049) are students of Third semester B.Tech in Computer Science & Engineering, JAIN (Deemed-to-be University), hereby declare that the project centric learning work titled "Decentralized P2P" has been carried out by us and submitted in partial fulfilment for the project centric learning activity of degree in Bachelor of Technology in Computer Science & Engineering during the academic year 2023-2027.

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CHAPTER 1

ACKNOWLEDGEMENT

Privora is a privacy-first, secure social network built to fundamentally reshape how users interact online. Rejecting the aggressive data-mining practices of modern social platforms, Privora combines end-to-end encrypted style messaging features, self-destructing data, and isolated user vaults to keep data strictly within user control.

The application serves a dual purpose: it operates as a fully functional social platform allowing for posts, discovery, and connections, while also doubling as a highly secure instant messenger capable of burning messages and QR keys immediately after consumption.

Problems Solved

- **Data Harvesting**: Traditional social media platforms mine user data. Privora treats user interactions as private cryptographic events.
- **Message Permanence:** Most applications store chats indefinitely. Privora implements a time-to-live (TTL) expiration, shredding messages securely after they are read.
- **Invasion of Privacy:** Sensitive texts like passwords or PINs are often left in chat logs. Privora solves this by offering "Secure Keys" (QR codes or passwords) that self-destruct the exact moment they are viewed.
- **Authentication Bloat**: The system uses local browser vaults to handle "passwordless-style" session identity, removing the need for complex and vulnerable OAuth integrations during this prototype phase.

CHAPTER 2

LITERATURE SURVEY

Paragraph:

Existing messaging platforms such as WhatsApp, Telegram, and Signal provide encrypted communication but still lack complete control over user data. These systems often store messages and metadata for long durations, creating privacy risks. Features like disappearing messages exist but are not fully secure, and sensitive data like passwords or OTPs can remain exposed. Additionally, traditional authentication methods increase system complexity and vulnerability. Privora addresses these issues by introducing a privacy-first approach with secure, self-destructing communication and simplified authentication.

Key Points:

- Existing apps collect **user data & metadata**
- Messages are stored **indefinitely**
- Disappearing messages are **not fully secure**
- Sensitive information remains **in chat history**
- Authentication systems are **complex & vulnerable**
- **Privora solutions:**
    - Cryptographic interactions
    - TTL-based self-destructing messages
    - Secure Keys (view-once secrets)
    - Local vault (passwordless authentication)

CHAPTER 3

OBJECTIVE AND METHODOLOGY

3.1 Objectives

The primary objectives of this project are:

- To design a **secure messaging platform** with minimal backend dependency
- To ensure **user privacy by avoiding storage of sensitive data**
- To implement a **temporary chat system** with optional retention
- To develop a **request-based communication system** for controlled interaction
- To provide a **simple and user-friendly interface**
- To prevent spam and unauthorized access
- To eliminate reliance on third-party authentication systems

3.2 Methodology

The methodology adopted for this project follows a structured approach:

1.Requirement Analysis

Requirement analysis is the first step in developing a secure messaging system. It focuses on identifying the limitations of existing systems, understanding privacy and security challenges like data breaches, and defining key requirements such as strong encryption, secure access, and reliable communication. It also ensures that the system meets user needs and provides a smooth user experience. Additionally, it helps in planning a scalable and efficient system for future improvements.

2.System Design

System design is the phase where the overall structure of the messaging system is planned. It involves designing a front-end focused architecture and defining the user flow, including login, connection, and chat processes. Security mechanisms are also planned to ensure data protection and safe communication. Additionally, this phase ensures the system is efficient, user-friendly, and scalable for future needs. It also helps in organizing different system components and their interactions effectively. Proper design reduces future errors and simplifies maintenance. It ensures that the system can handle increasing users and data efficiently.

3. Implementation

Implementation is the stage where the actual development of the system takes place. It involves developing user interface components and implementing a secure authentication system. A request-based connection model is designed, along with temporary chat functionality for communication. This phase ensures that all features work smoothly and meet the defined requirements.

4. Testing

Testing is the phase where the system is evaluated to ensure everything works correctly. It includes functional testing of all features, security testing to check safe data handling, and performance testing to ensure fast and smooth responsiveness. This phase helps in identifying and fixing errors before deployment.

5. Deployment

Deployment is the final phase where the system is prepared for release. It involves finalizing the system design, preparing proper documentation, and ensuring usability and stability. This phase ensures the system is ready for real-world use and provides a smooth experience to users.

CHAPTER 4

1.Technology Stack

Privora utilizes a decoupled, modern Client-Server Architecture.

Frontend (Client)

- **Framework** : React 19 / Vite. The absolute cutting-edge of the React ecosystem, offering lightning-fast Hot Module Replacements.
- **Language** : TypeScript. Brings strict type safety across network boundaries.
- **Styling :** Tailwind CSS v4. Enables highly responsive, modern glassmorphed user interfaces.
- **State Management :** Zustand. A lightweight, robust state container that easily persists complex vaults to browser local storage.
- **Icons & Extras :** `lucide-react` for crisp SVG icons and `react-qr-code` for real-time cryptographic image generation.

Backend (Server)

- **Engine:** Node.js processing over Express.js.
- **Database :** better-sqlite3`. A synchronous, ultra-fast C++ driven SQLite engine ideal for high-throughput messaging relays.
- **Security :** `bcrypt` for one-way password hashing and `uuid` for generating unguessable UUIDv4 primary keys.

2 .Key Features Implemented

Security & Privacy

1. Disappearing Messages : A global timer shreds chat messages automatically after the recipient views them.

2. Secure Keys (Burn After Reading) : The ability to send highly classified text (like crypto addresses or passwords) as a QR Code. The interface forces the recipient to click to reveal, automatically destroying the key immediately upon closure.

3. End-to-End Style Encryption : Chat messages are treated as `payloads` in the backend database. The backend acts solely as a ; Relay handing the undecipherable blocks of JSON back to the target client.

CHAPTER 5

SYSTEM ARCHITECTURE & WORKFLOW

The software architecture resolves around stateless REST requests:

1. Polling Engine : The frontend utilizes an incredibly lightweight background poller (`hydrateFromBackend`) that consistently pulls the latest JSON patches for Connections, Posts, and new Messages at scheduled intervals without flooding the user's browser.

2. Payload Protection : When Alice sends Bob a file (under 2MB), the frontend maps it to a `Base64` data string, packages it into a strongly-typed `Message` JSON object, and sends it to `POST /messages`.

3. Asynchronous Resolution : SQLite rapidly evaluates the `toUserId` matrix to locate Bob's new messages upon his next polling interval.

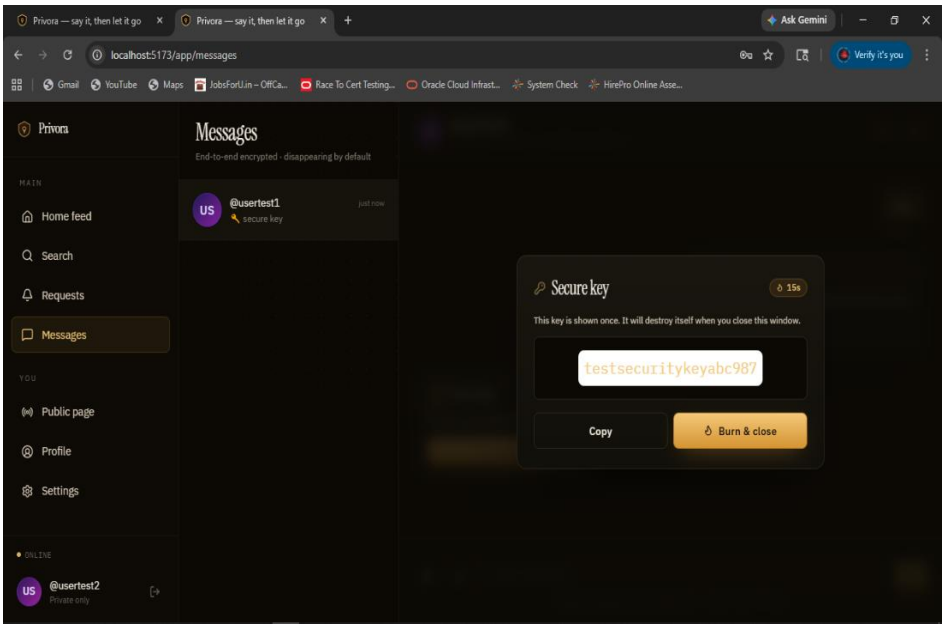

Fig 1.1(Revealing of security key in user2 chat)

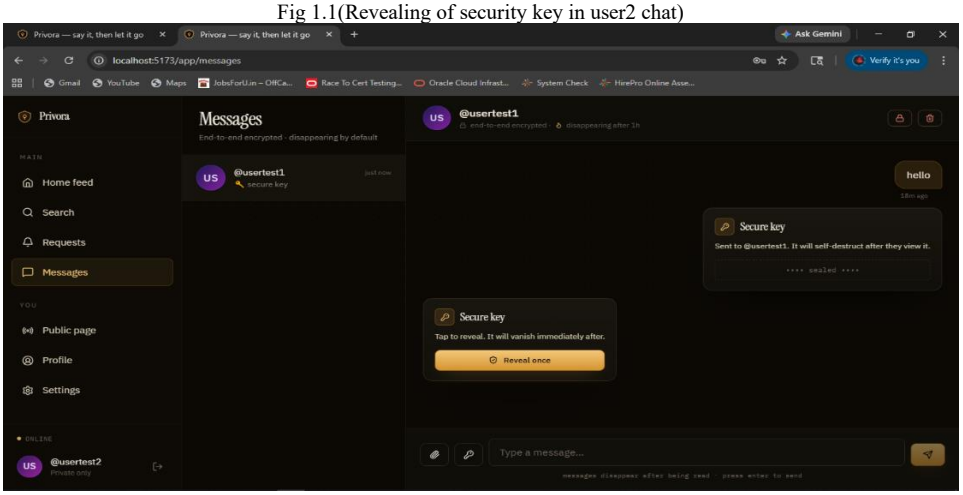

Fig1.2 (Receiving security key from usrr1)

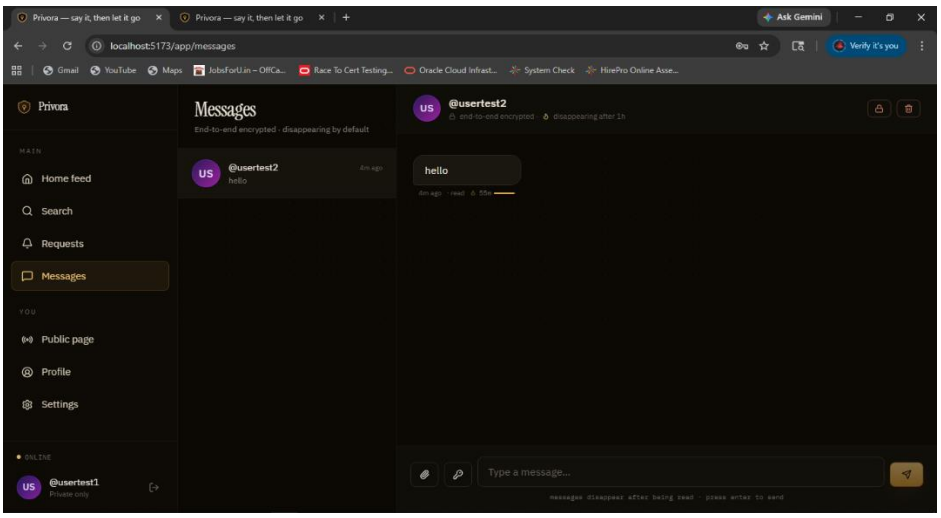

Fig1.3 (Cross E2EE messaging between users)

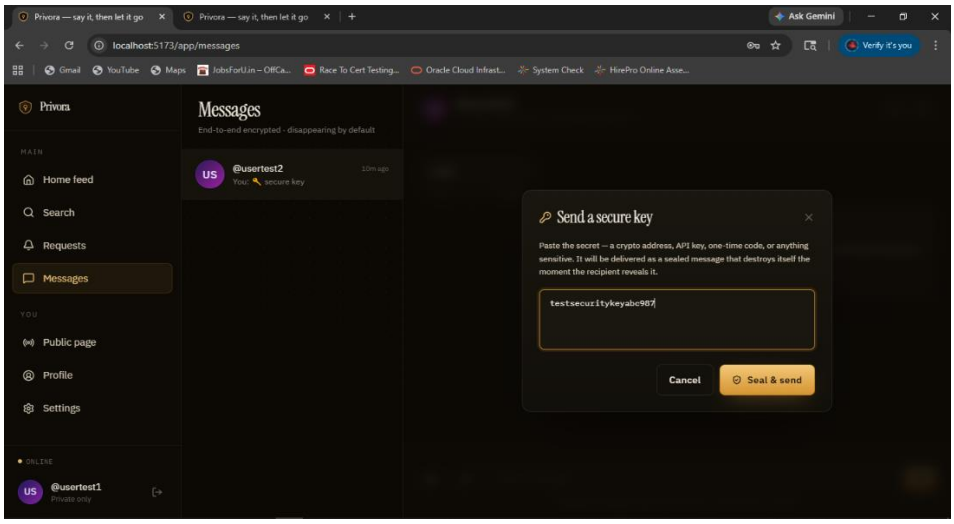

Fig1.4(sending security from user1 to user2)

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